



**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING  
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)**

**Academic Year:** 2025-26

**Semester:** VI

**Class / Div / Branch:** T.E./ B1/ CSE(AI&ML)

**Subject:** SEPM Lab

**Name of Instructor:** Shruthi Nair

**Name of Student:** Prajwal Dhanawade

**Student ID:** 24206007

**Roll No:** 75

**Date of Performance:**

**Date of Submission:**

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**Experiment No. 2**

**Aim:** To understand Version Control System / Source Code Management, install git and create a GitHub account.

**Lab Outcomes:** To understand the fundamentals of DevOps engineering and be fully proficient with DevOps terminologies, concepts, benefits, and deployment options to meet your business requirements.

**Theory:** Version Control System:

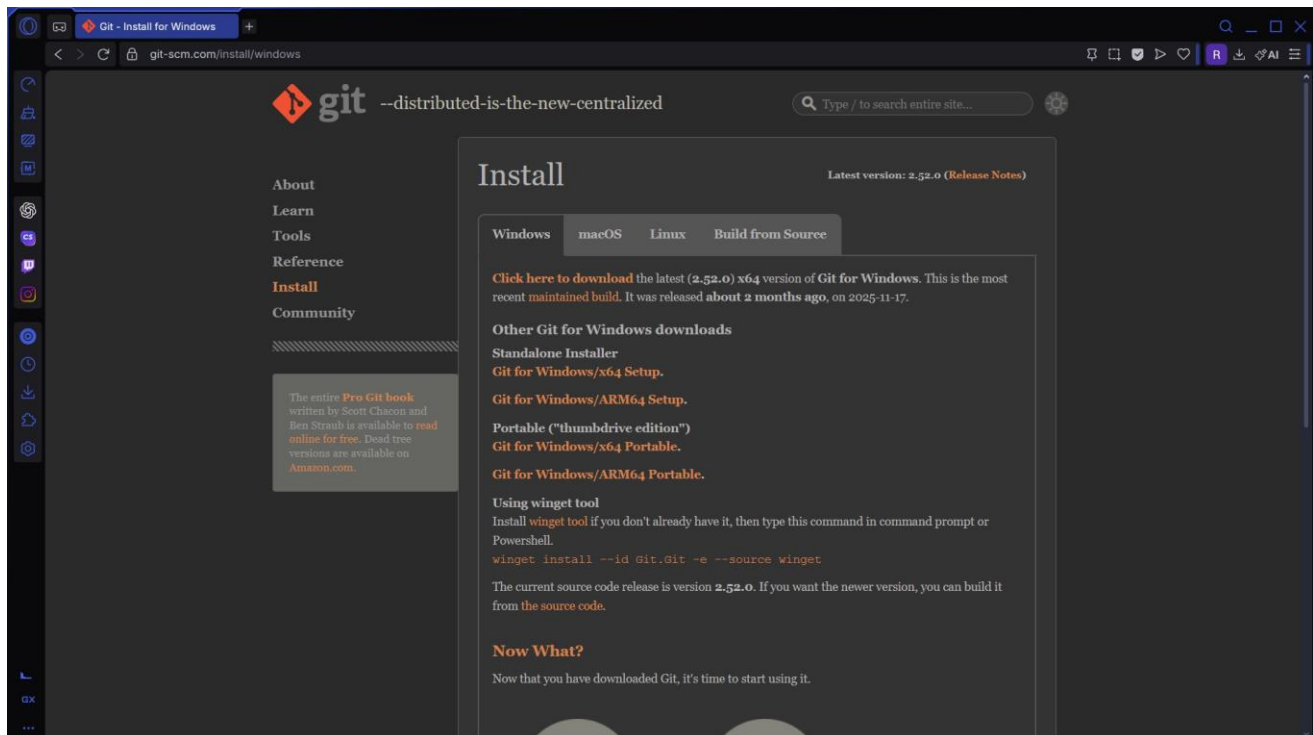
Version control, also known as source control, is the practice of tracking and managing changes to software code. Version control systems are software tools that help software teams manage changes to source code over time. As development environments have accelerated, version control systems help software teams work faster and smarter. They are especially useful for DevOps teams since they help them to reduce development time and increase successful deployments. Version control software keeps track of every modification to the code in a special kind of database. If a mistake is made, developers can turn back the clock and compare earlier versions of the code to help fix the mistake while minimizing disruption to all team members. Version control helps teams solve these kinds of problems, tracking every individual change by each contributor and helping prevent concurrent work from conflicting. Changes made in one part of the software can be incompatible with those made by another developer working at the same time. This problem should be discovered and solved in an orderly manner without blocking the work of the rest of the team. Further, in all software development, any change can introduce new bugs on its own and new software can't be trusted until it's tested. So testing and development proceed together until a new version is ready.



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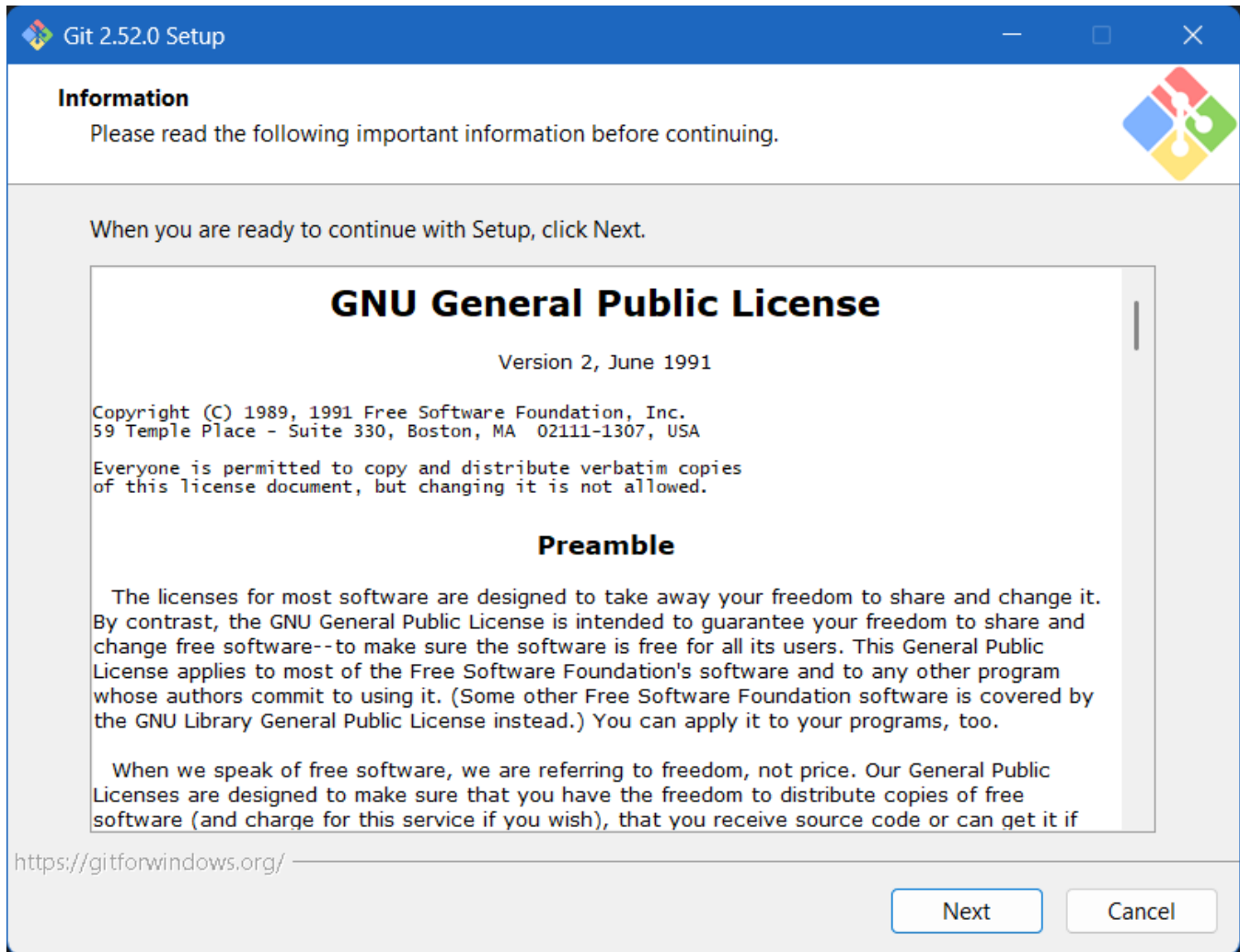
### Output:

### Git Installation





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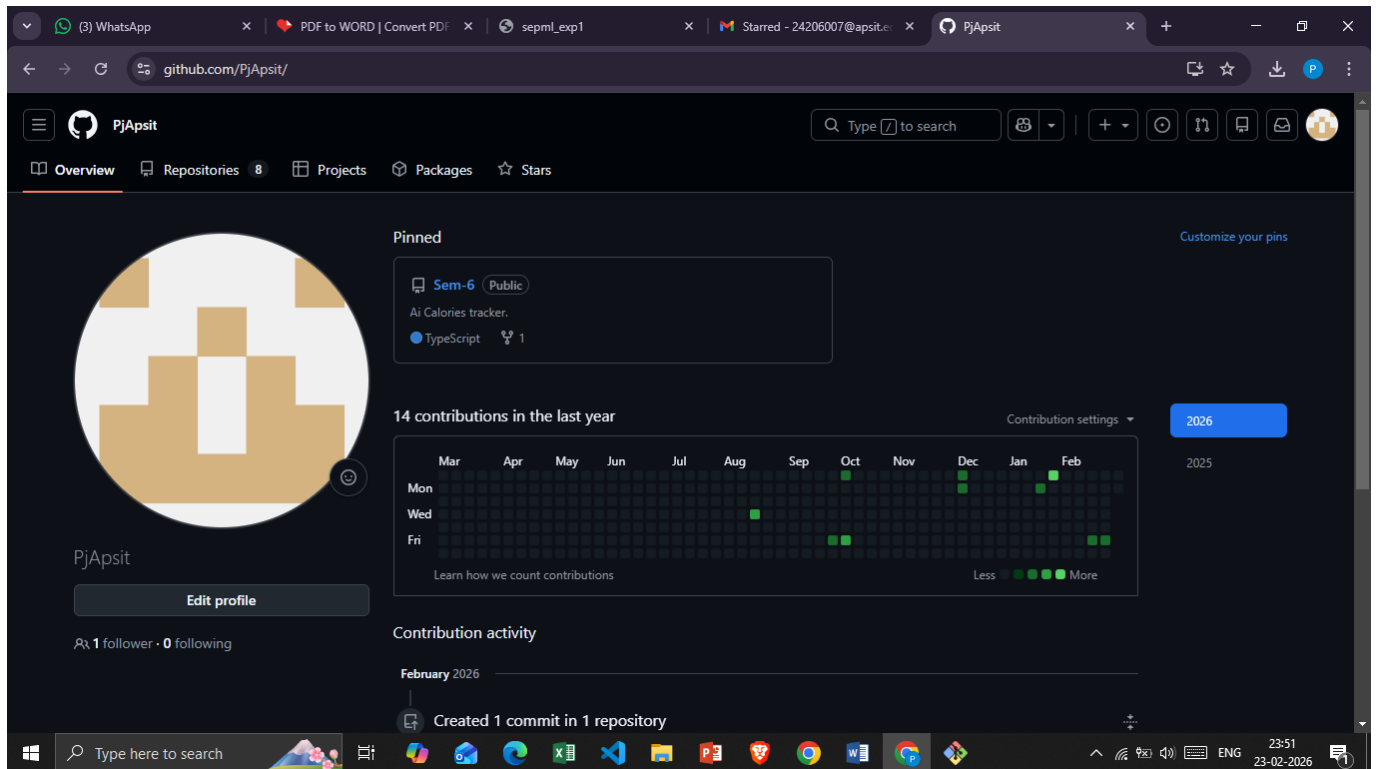
```
MINGW64:/c/Users/HP
HP@Prajwal MINGW64 ~
$ Prajwal Here (* *)
```



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**Conclusion:** We studied the basic concept of git and version control system and also how to install git and create account on Git hub.